US 20250268125A1

(19) United States
(12) Patent Application Publication
Johnson et al.

(10) Pub. No.: US 2025/0268125 A1
(43) Pub. Date: Aug. 28, 2025

(54) MULTI-MOTOR COMMON DRIVE SYSTEM FOR SEED METERS

(71) Applicant: **Great Plains Manufacturing, Inc.**, Salina, KS (US)

(72) Inventors: **Gerald R. Johnson**, Newton, KS (US); **Donald H. Riffel**, Smolan, KS (US)

(21) Appl. No.: **18/584,762**

(22) Filed: **Feb. 22, 2024**

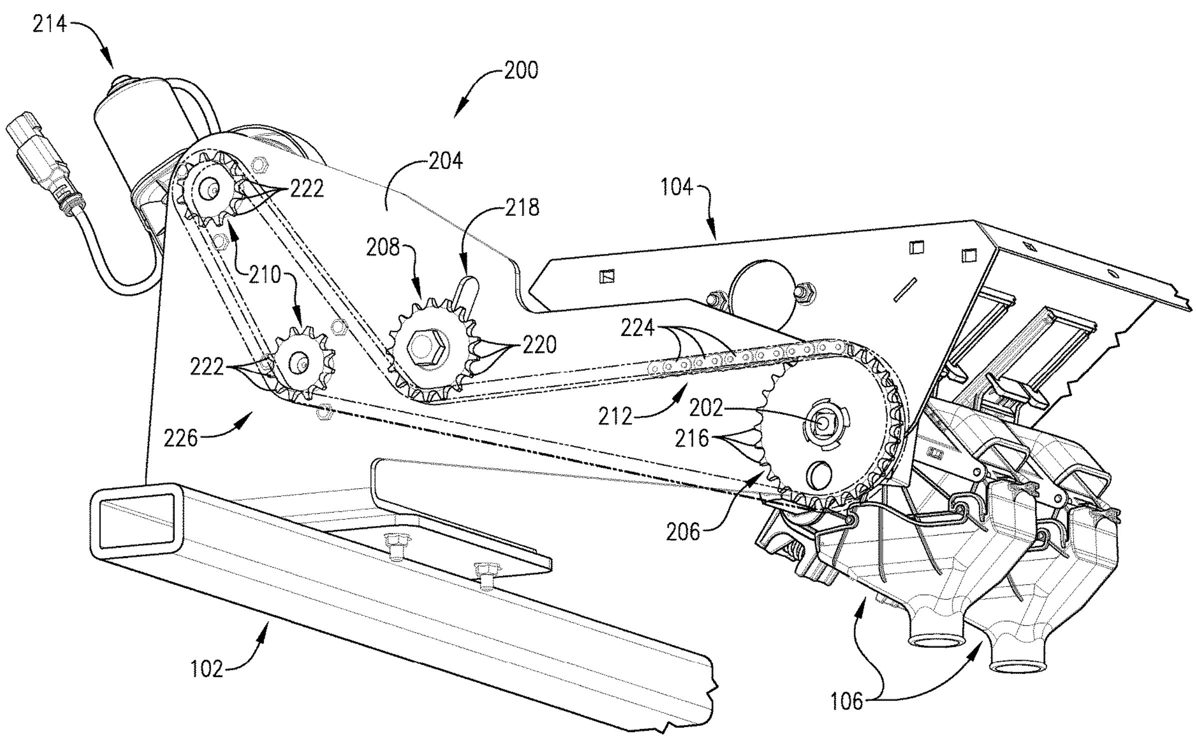
Publication Classification

(51) **Int. Cl.**
A01C 19/02 (2006.01)
A01C 7/12 (2006.01)

(52) **U.S. Cl.**
CPC ***A01C 19/02*** (2013.01); *A01C 7/12* (2013.01)

(57) **ABSTRACT**

A drive system for delivering a plurality of seeds into a number of furrows via a number of seed meters. The drive system comprises a meter shaft drivably connectable to the seed meters and a number of electric motors drivably connected to the meter shaft for actuating the seed meters via the meter shaft.



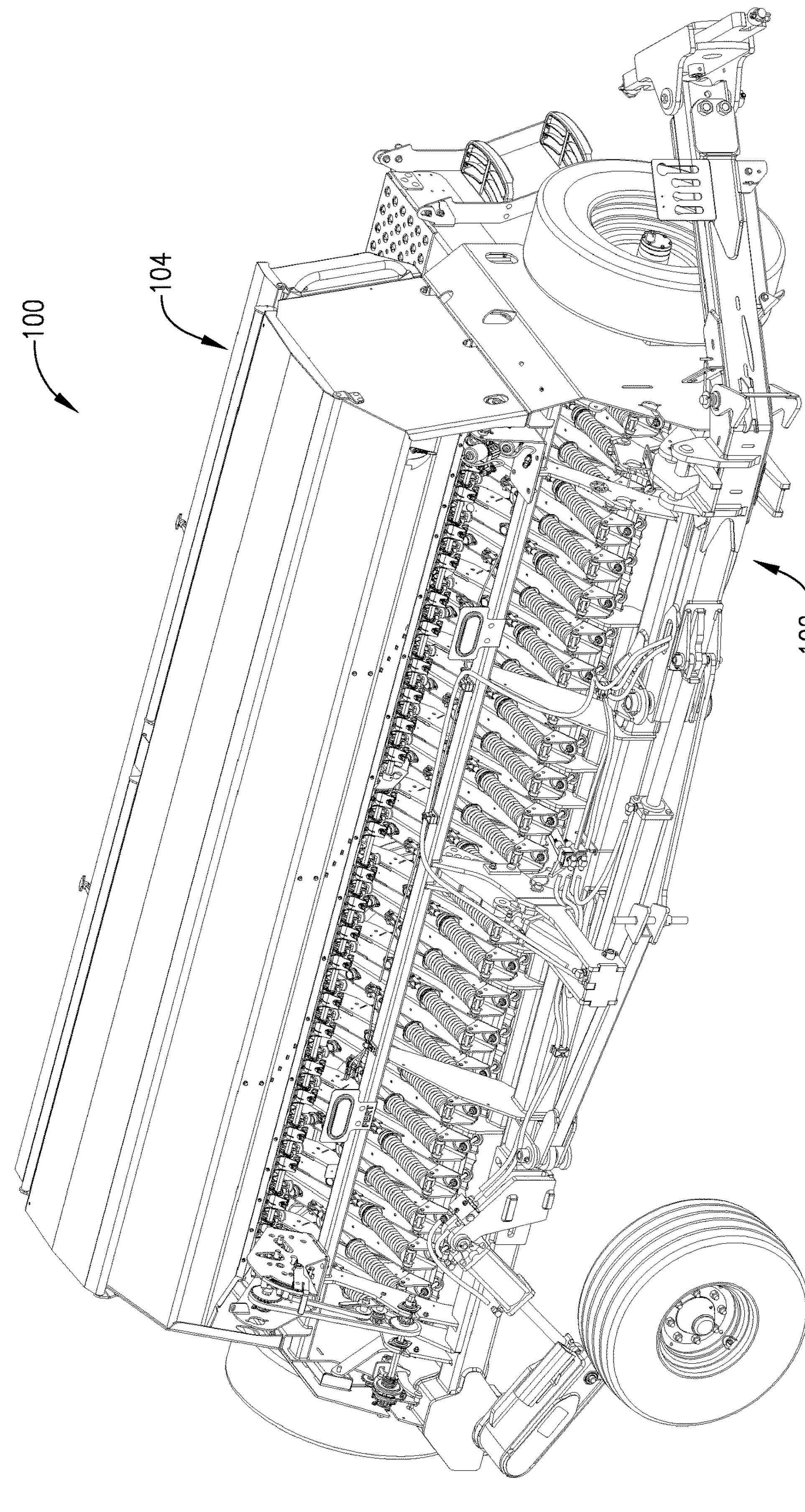


FIG. 1

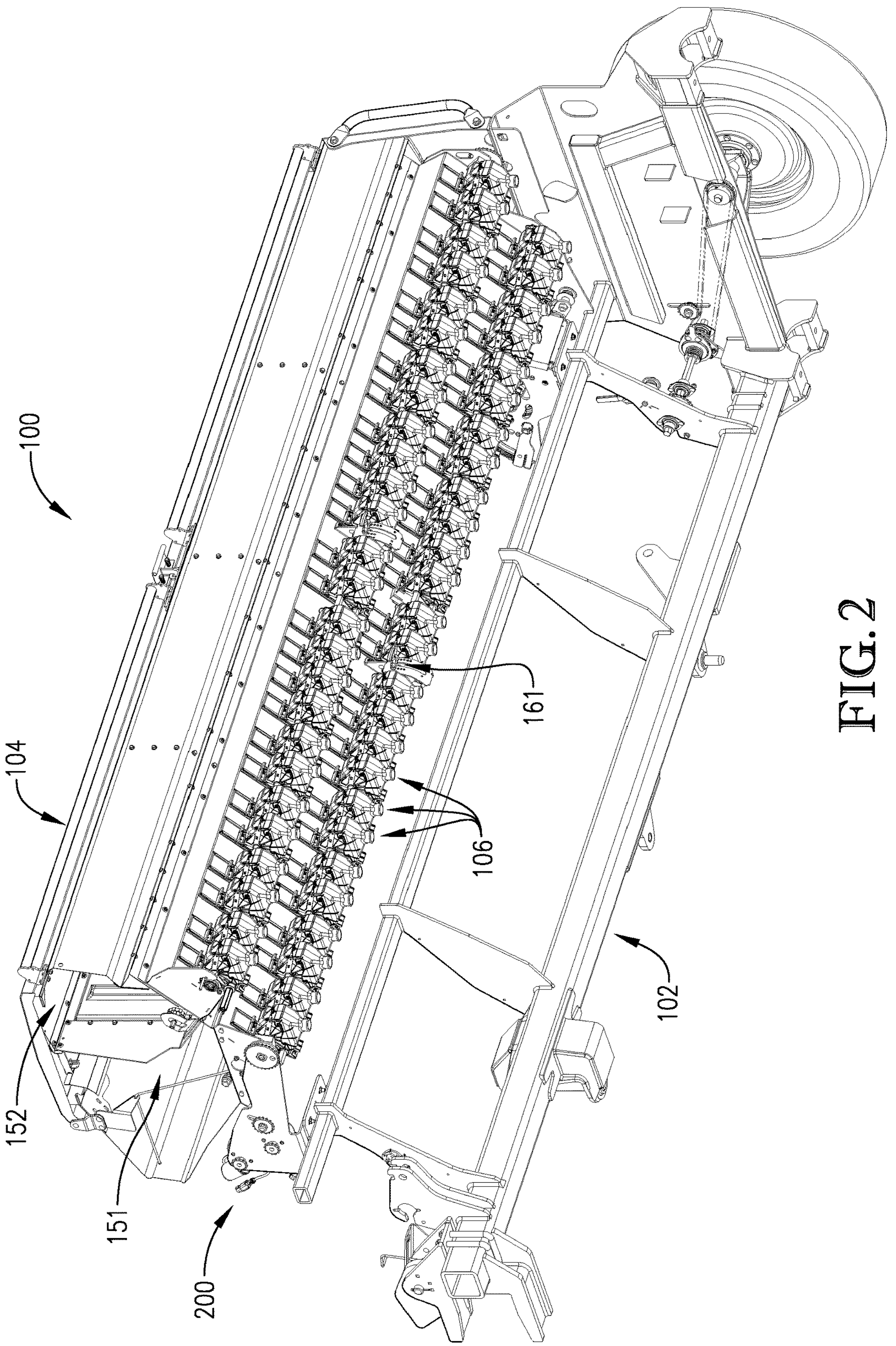


FIG. 2

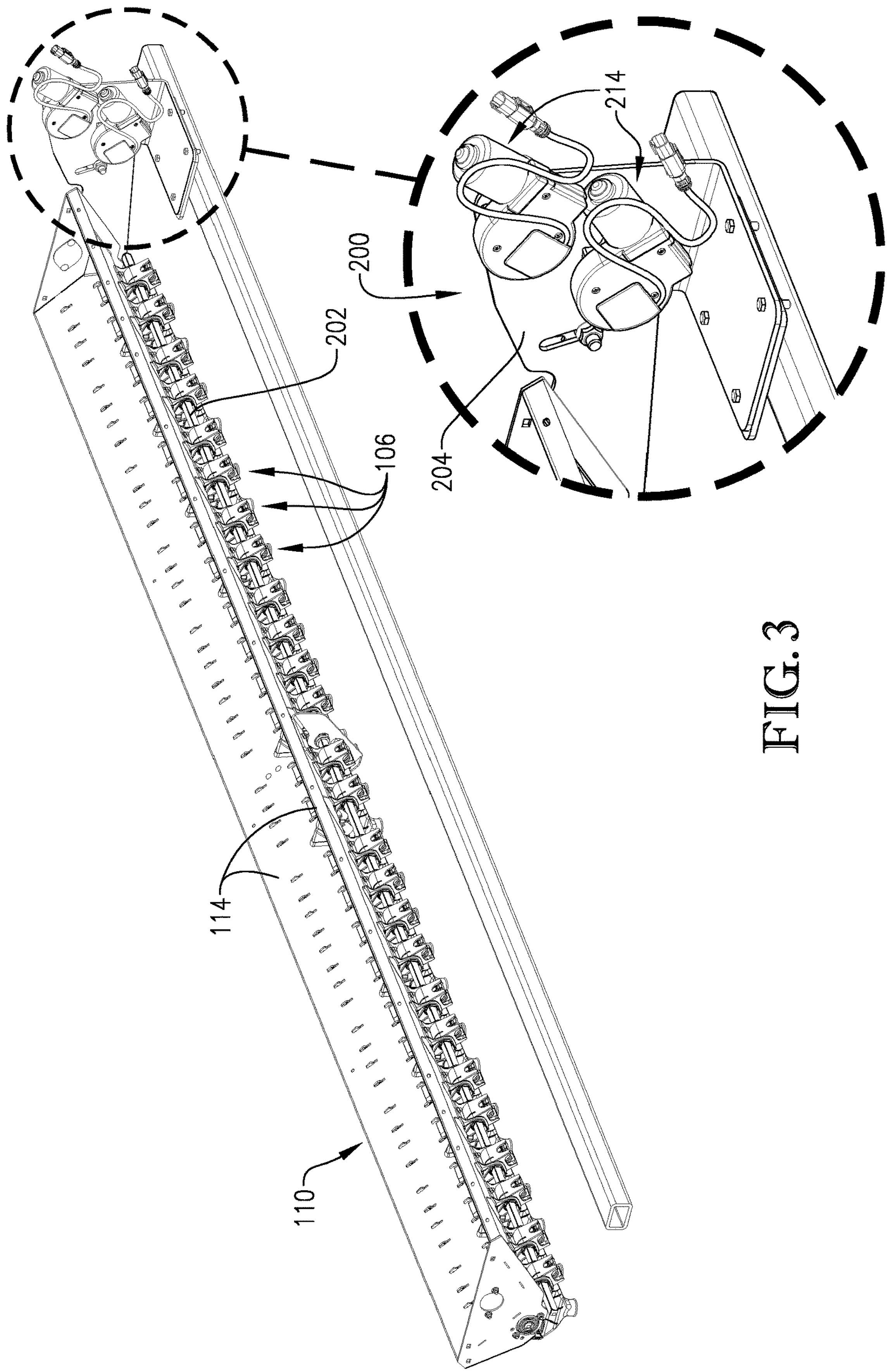


FIG. 3

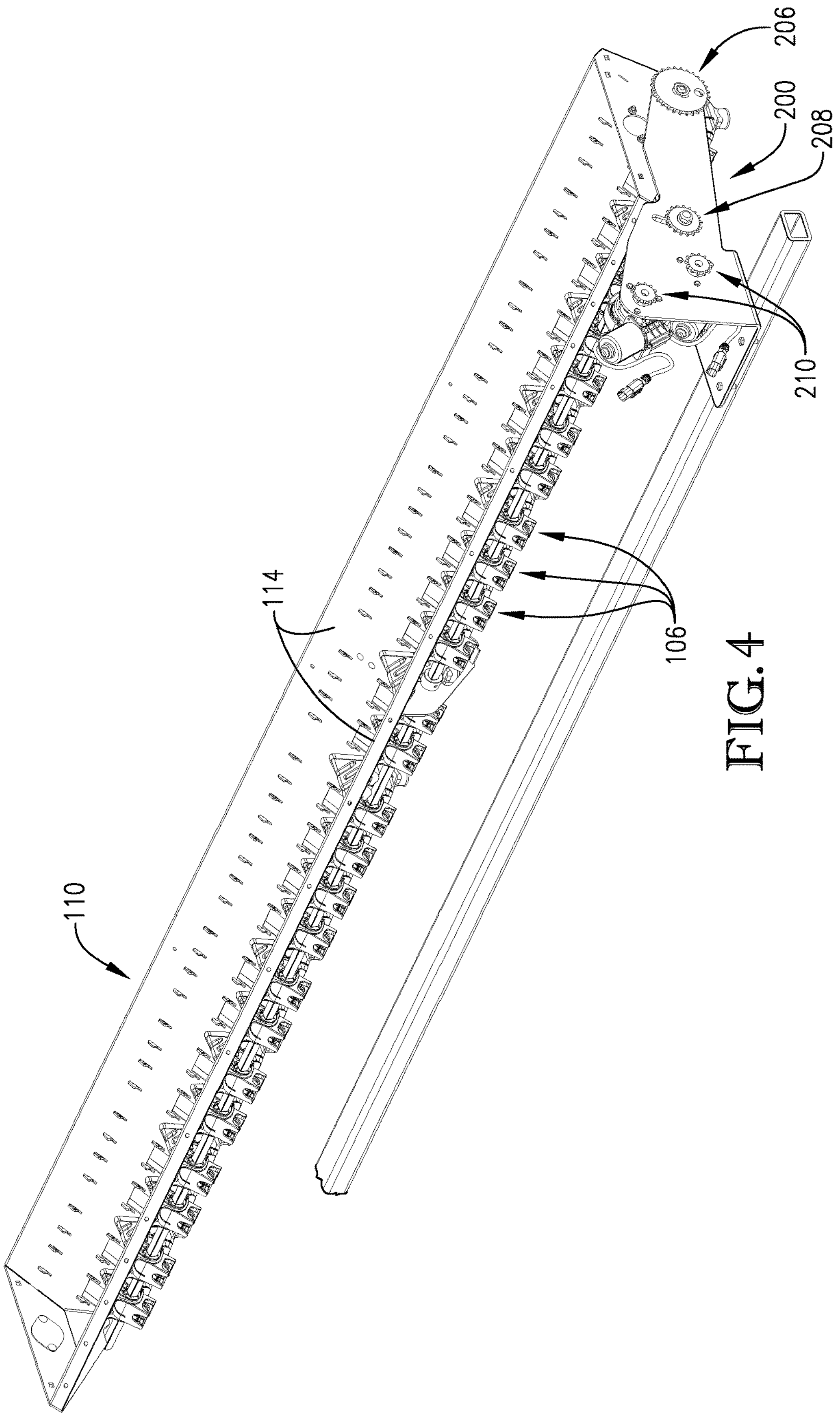


FIG. 4

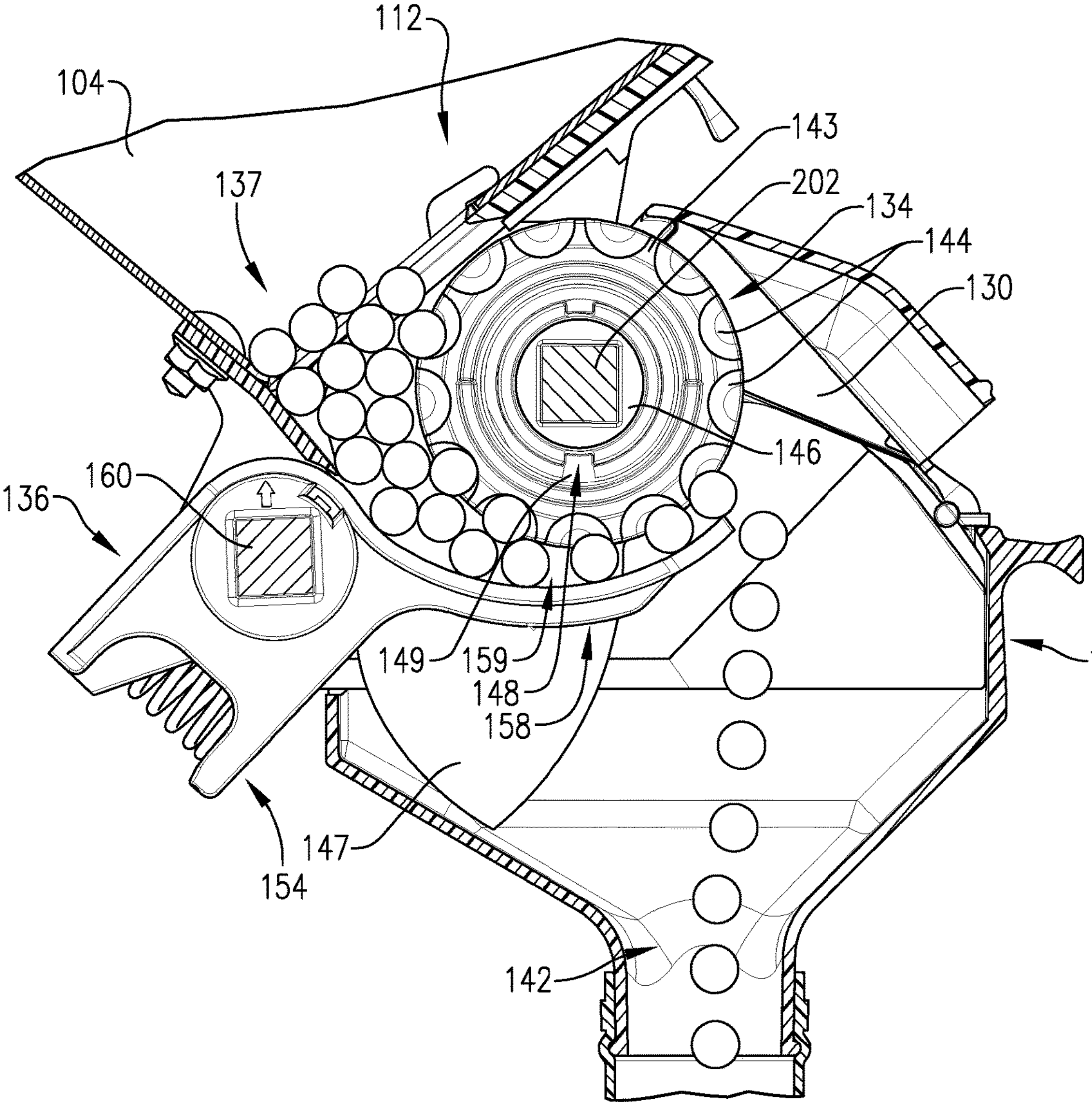

112
104
137
143
202
134
144
130
146
160
136
132
149
159
148
158
154
147
142

FIG. 5

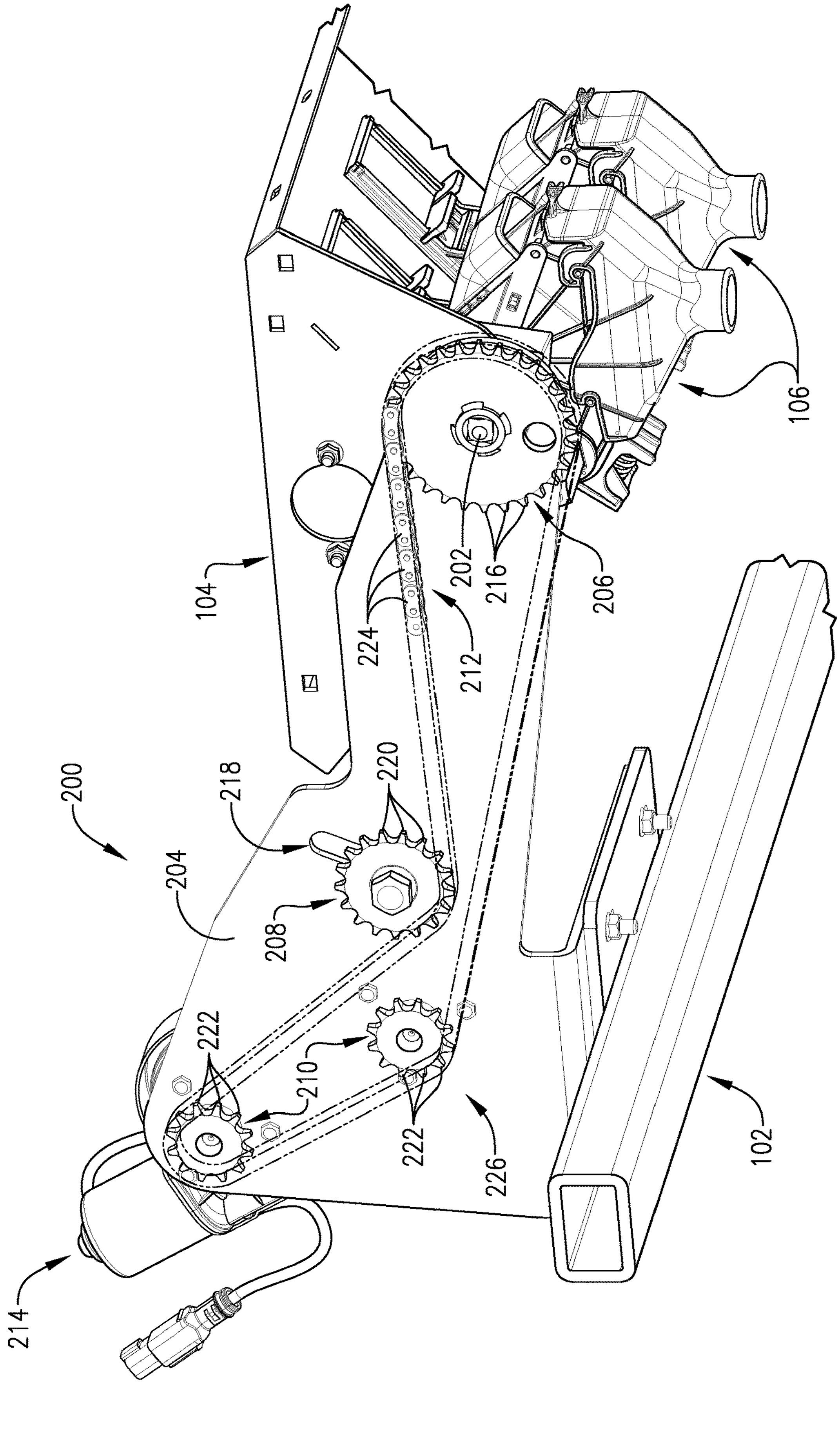

106
104
202
216
206
224
212
200
218
220
204
208
222
210
222
226
102
214

FIG. 6

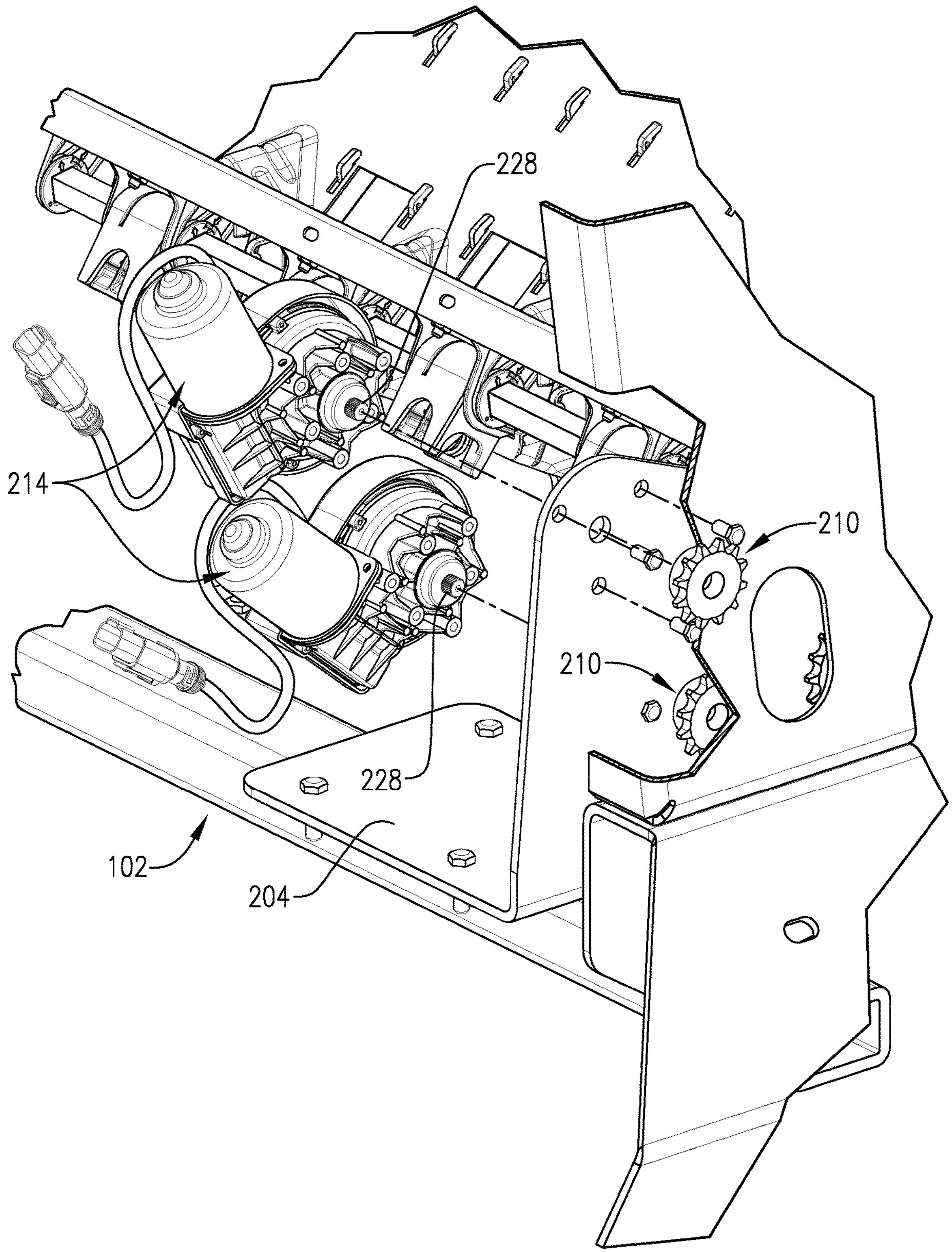


FIG. 7

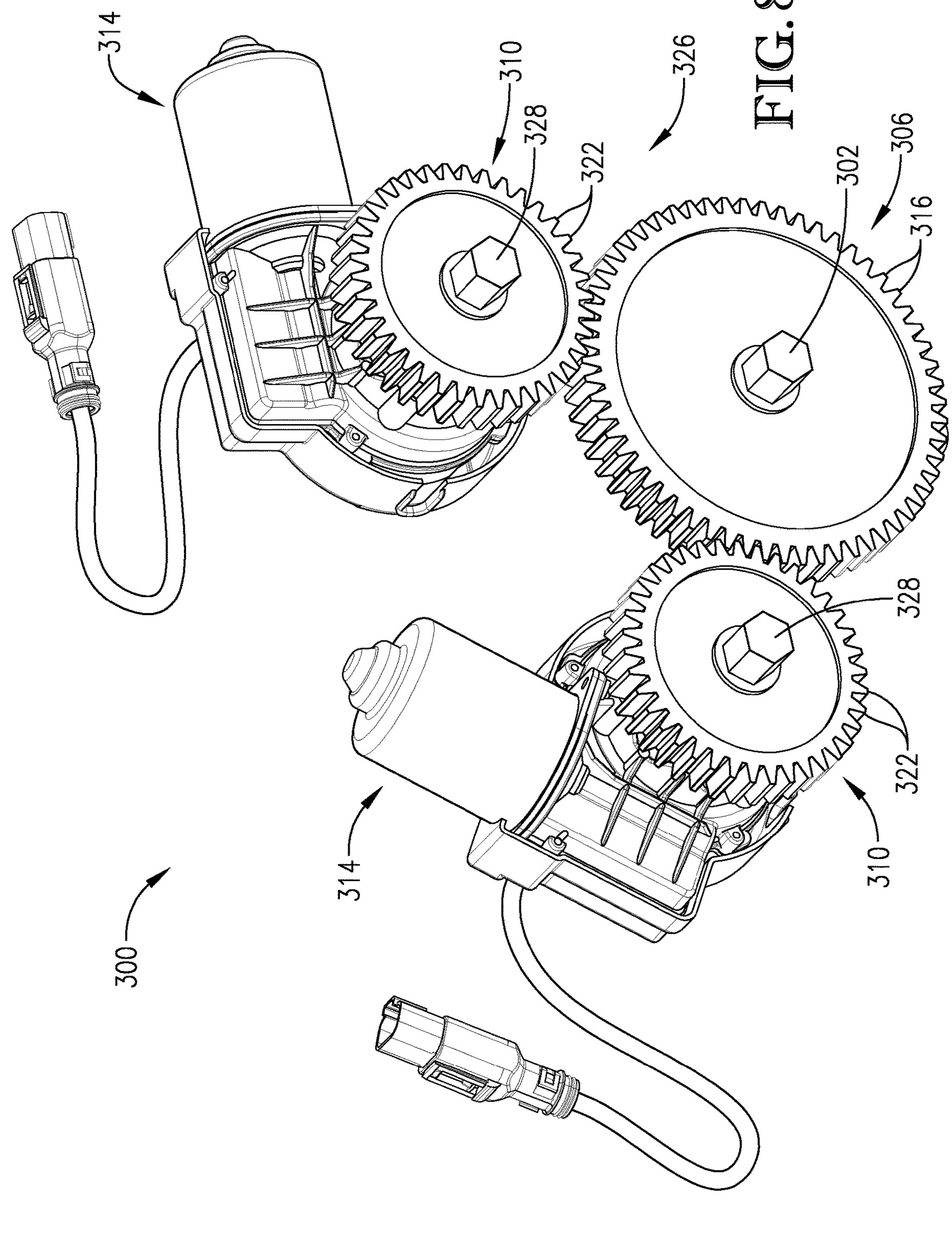


FIG. 8

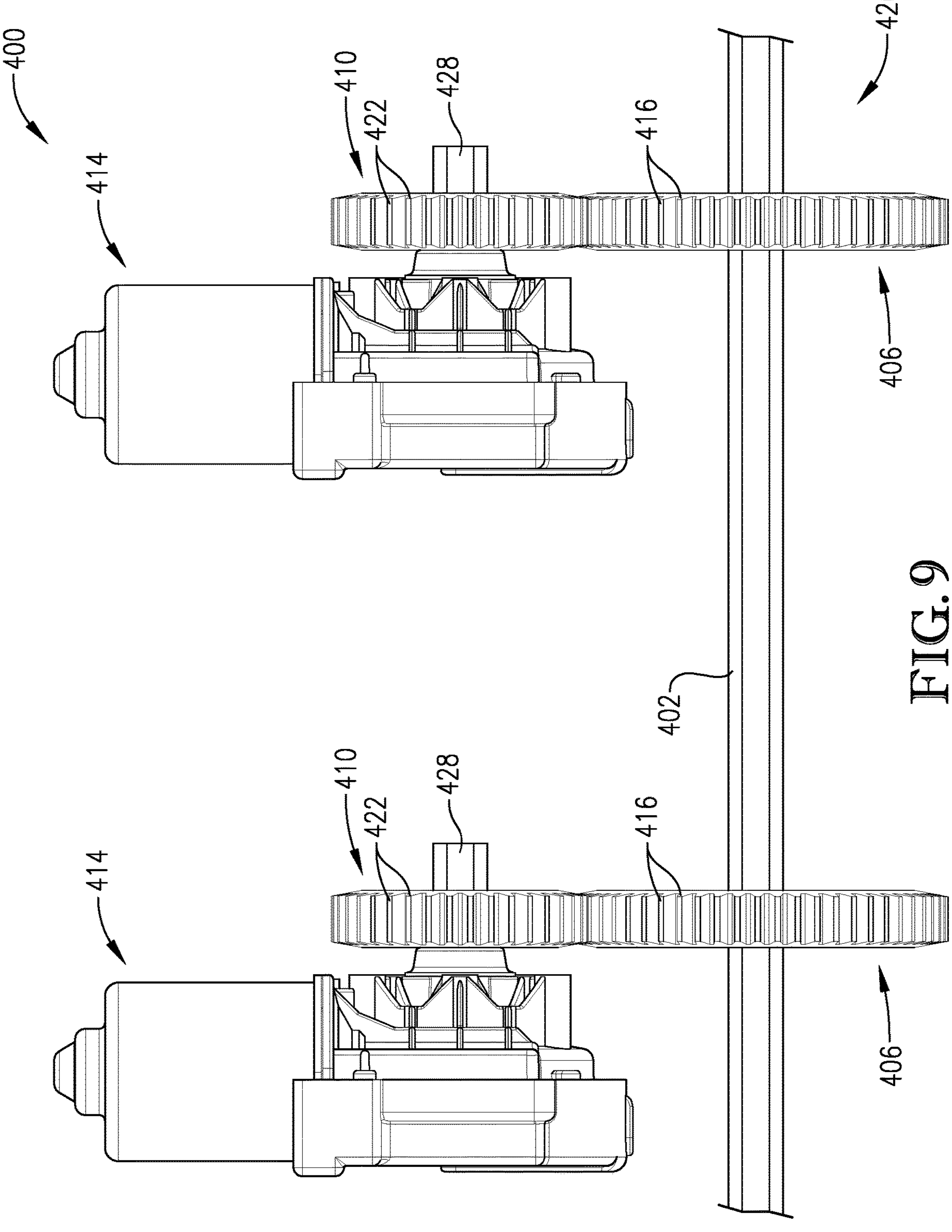


FIG. 9

MULTI-MOTOR COMMON DRIVE SYSTEM FOR SEED METERS

FIELD OF THE INVENTION

[0001] Embodiments of the present invention are directed generally to agricultural implements for seeding. In more detail, embodiments of the present invention are directed to agricultural implements that include seed bins and seed metering devices for dispensing seed and other agricultural products.

BACKGROUND OF THE INVENTION

[0002] Certain agricultural implements, such as seed drills are configured to dispense agricultural products (e.g., seed and/or treatment) into or onto the ground. Commonly, a seed drill will include one or more bins that hold agricultural product. As the seed drill is pulled through a field by a tractor or other prime mover, agricultural product can be dispensed from the bins via a plurality of metering devices associated with the bins.

[0003] Metering devices of agricultural implements are often powered by a ground drive (i.e., movement of the agricultural implements imparts movement of the metering devices) or by an electronic control unit (ECU) controlled hydraulic drive. Ground drives are convenient, but not very versatile because the travel speed of the agricultural implements across the ground dictates the metering speed of the metering devices. ECU controlled hydraulic drives introduce metering control independent of ground speed, but are undesirably complex.

[0004] An electric motor drive is simple and provides electronic metering control independent of ground speed, but powering several metering devices requires a large RPM range. A large, expensive motor is needed for the upper end of the RPM range but is oversized in some stages of operation. Furthermore, using several small electric motors to drive individual metering devices is undesirable because additional drive trains will be required and the additional electric motors introduces several potential points of failure.

SUMMARY OF THE INVENTION

[0005] In one embodiment of the present invention, there is provided a drive system for delivering seeds into a number of furrows via a number of seed meters. The drive system comprises a meter shaft drivably connectable to the seed meters. The drive system further includes a number of electric motors drivably connected to the meter shaft for actuating the seed meters via the meter shaft.

[0006] In another embodiment of the present invention, there is provided a seed drill comprising a drive system for delivering seeds into a number of furrows and a control system for controlling the drive system. The drive system includes a number of seed meters, a meter shaft, and a number of electric motors. Each seed meter is configured to deliver some of the seeds into one of the furrows. The meter shaft is drivably connected to the seed meters. The electric motors are drivably connected to the meter shaft for actuating the seed meters via the meter shaft. The electric motors include a first electric motor operating as a lead motor and a second electric motor operating as a follower motor. The control system is communicatively connected to the first electric motor and is configured to transmit a command signal to the first electric motor for activating the first electric motor to achieve a first operating state.

[0007] In a further embodiment of the present invention, there is provided a seed drill comprising a drive system for delivering a number of seeds into a number of furrows and a control system for controlling the drive system. The drive system includes a number of seed meters, a meter shaft, a number of electric motors, and a drive train. Each seed meter is configured to deliver some of the seeds into one of the furrows. The meter shaft is drivably connected to the seed meters. The electric motors are drivably connected to the meter shaft for actuating the seed meters via the meter shaft. The electric motors include a first electric motor operating as a leader motor and a second electric motor operating as a follower motor. The second electric motor is configured to be switched to operating as a leader motor. The electric motors have different operating specifications from each other. The drive train drivably connects the first electric motor and the second electric motor to the meter shaft at one axial location on the meter shaft. The control system is communicatively connected to the first electric motor and configured to transmit a command signal via pulse width modulation (PWM) control area network (CAN) communication or a similar communication scheme to the first electric motor for activating the first electric motor to achieve a first operating state. The second electric motor is synchronized with the first electric motor to achieve a second operating state substantially equal to the first operating state. The first and second operating states are at least one of a speed, a torque, an electric current (e.g., an electric current input or consumption at one of the first and second electric motors or an electric current output from the controller), voltage, and a direction of rotation.

[0008] A further embodiment of the invention is a method of planting seeds. The method comprises steps of (a) connecting a common meter shaft to a plurality of individual seed meters so that rotation of the common meter shaft causes rotation of the plurality of individual seed meters; (b) driving rotation of the common meter shaft in a first operating state where n individual electric motors power rotation of the common meter shaft; and (c) driving rotation of the common meter shaft in a second operating state where m individual electric motors power rotation of the common meter shaft. The number of individual electric motors m is at least one more than n.

[0009] This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the detailed description. This summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Other aspects and advantages of the present invention will be apparent from the following detailed description of the embodiments and the accompanying drawing figures.

BRIEF DESCRIPTION OF THE FIGURES

[0010] Embodiments of the present invention are described herein with reference to the following drawing figures, wherein:

[0011] FIG. **1** is a top perspective view of an agricultural implement according to embodiments of the present invention;

[0012] FIG. **2** is a bottom perspective view of portions of the agricultural implement of FIG. **1**;

[0013] FIG. **3** is a top perspective view of certain components of the agricultural implement of FIG. **1**;

[0014] FIG. **4** is another top perspective view of certain components of the agricultural implement of FIG. **1**;

[0015] FIG. **5** is a cutaway side elevation view of certain components of the agricultural implement of FIG. **1**;

[0016] FIG. **6** is a bottom perspective view of a drive system of the agricultural implement of FIG. **1**;

[0017] FIG. **7** is a top perspective view of certain components of the drive system of FIG. **6**;

[0018] FIG. **8** Is a top perspective view of a drive system constructed according to another embodiment of the invention; and

[0019] FIG. **9** is a top perspective view of a drive system constructed according to another embodiment of the invention.

[0020] The drawing figures do not limit the present invention to the specific embodiments disclosed and described herein. The drawings are not necessarily to scale, emphasis instead being placed upon clearly illustrating the principles of the invention.

DETAILED DESCRIPTION

[0021] The following detailed description of the present invention references various embodiments. The embodiments are intended to describe aspects of the invention in sufficient detail to enable those skilled in the art to practice the invention. Other embodiments can be utilized and changes can be made without departing from the scope of the present invention. The following detailed description is, therefore, not to be taken in a limiting sense. The scope of the present invention is defined only by the appended claims, along with the full scope of equivalents to which such claims are entitled.

[0022] Embodiments of the present invention are directed generally to agricultural implements, such as seed and/or treatment drill **100** illustrated in FIG. **1**. The drill **100** may comprise a frame **102** that is towed by a tractor or other prime mover (not shown). The frame **102** may support a bin **104** that extends laterally across the frame **102**. As will be described in more detail below, the bin **104** is configured to hold agricultural products, such as seed and/or treatment (e.g., fertilizer, pesticides, etc.), for dispensing into and/or onto the ground. In some embodiments, the bin **104** may be divided into multiple containment sections that are each configured to hold distinct types of agricultural product. For instance, the bin **104** may include a forward containment section **151** and a rearward containment section **152**. As such, the forward containment section **151** may be configured to hold a first type of agricultural product (e.g., seed), while the rearward containment section **152** may be configured to hold a second type of agricultural product (e.g., treatment). As used herein, the term "treatment" may refer to fertilizers, pesticides, herbicides, nutrients, or other additives used in connection with seeding.

[0023] To facilitate dispensing of agricultural product, the drill **100** may additionally comprise a plurality of metering devices **106** secured to a bottom side of the bin **104**, such as illustrated in FIG. **2**. In instances in which the bin **104** is divided into multiple containment sections **151**, **152**, embodiments may provide for a plurality of metering devices **106** to be secured to a bottom of each containment section. The metering devices **106** (lower row in FIG. **2**) associated with containment section **151** are driven via a drive system **200**, which will be described in more detail below. Metering devices (upper row in FIG. **2**) associated with containment section **152** may be driven via a conventional drives system such as a ground drive or electronic control unit (ECU) hydraulic drive (which will not be described further). Alternatively, the metering devices associated with containment section **152** may be driven by a drive system similar to drive system **200**.

[0024] The bin **104** may comprise a generally rectangular container configured to hold agricultural products (e.g., seed and/or treatment). Turning to FIGS. **3** and **4**, a bottom section **110** of the bin **104** may be formed in a triangular or funnel shape. In some embodiments, the bin **104** may include an individual bottom section **110** associated with each containment section **151**, **152**. The bottom section **110** may include a plurality of product openings **112** (FIG. **5**).

[0025] Agricultural product held within the bin **104** will be funneled downward under the force of gravity towards the product openings **112**, such that the agricultural product can pass through the product openings **112** to the metering devices **106** that function to dispense the agricultural product from the bin **104** into or onto the ground. To promote efficient funneling and mixing of the agricultural product within the bin **104**, some embodiments may provide for the bin **104** to include a mixing assembly (not shown) extending through the interior space of the bin **104**. Such mixing assemblies may include a rotatable shaft that extends through a length of the bin **104** and includes a plurality of mixing arms extending therefrom. Embodiments may provide for the rotatable shaft to be rotated in different directions and/or at different rotational speeds. Alternatively, the rotatable shaft may rotate at a generally constant speed and direction. Regardless, rotation of the rotatable shaft will cause the mixing arms to mix the agricultural product within the bin **104** so as to ensure proper mixing, funneling, and flow of the agricultural product through the bin **104**. In some additional embodiments, oscillators and/or agitators may be used to ensure proper mixing, funneling, and flow of the agricultural product through the bin **104**.

[0026] Turning now to the metering devices **106** in more detail, as illustrated in FIG. **5**, each metering device **106** may comprise an upper housing **130**, a lower housing **132** removably engaged with the upper housing **130**, a metering assembly **134** removably secured within an interior of the upper and/or lower housing **130**, **132**, and a gate assembly **136** removably secured within an interior of the upper and/or lower housing **130**, **132**. In operation of the drill **100**, a plurality of the metering devices **106** may be secured to an exterior bottom surface of the bin **104**. Thus, agricultural product can be passed from the bin **104** to the metering device **106**, such that the metering assembly **134** can convey the agricultural product through the metering device **106** and out of the metering device **106** where the agricultural product is dispensed into or onto the ground.

[0027] The upper housing **130** may present a product inlet **137** through selective opening of product doors (not shown) slidingly engaged with the top end of the upper housing **130**. Each of the product doors can be slidingly actuated between a closed position and an open position. In a closed position, the product doors restrict agricultural product from entering the metering device **106**. In contrast, in an open position, the product doors permit agricultural product to enter the metering device **106** via the presented product inlet **137**. In some embodiments, each of the product doors can be partially opened to various levels of extension (i.e., positions between completely closed and completely open) so as to regulate how much agricultural product can be provided into the metering device **106**.

[0028] As will be described in more detail below, the interior space of the metering device **106** may be divided into two agricultural product sections (e.g., a first agricultural product section and a second agricultural product section), such that two different types of agricultural product can be separately processed through the metering device **106**. To facilitate such separate processing, the product doors can be individually actuated. For instance, in order to supply a first agricultural product from the bin **104** to the metering device **106**, a first one of the product doors can be opened and a second one of the product doors can be closed so that the first agricultural product can be passed from the bin **104** to the interior space of the metering device **106** via the product unlet **137** presented by the opened first product door. In contrast, to supply a second agricultural product from the bin **104** to the metering device **106**, the second product door can be opened, and the first product door can be closed. As such, the second agricultural product can be passed from the bin **104** to the interior space of the metering device **106** via the product inlet **137** presented by the opened second product door.

[0029] Upon agricultural product being received into the upper housing **130** of the metering device **106**, via the product inlet **137** presented by either the first product door or second product door, the agricultural product can be conveyed through metering device **106**, via the metering assembly **134**. By way of such conveyance, the agricultural product can be dispensed from the lower housing **132** via a product outlet **142**. As illustrated in the drawings, the lower housing **132** may be shaped generally as a funnel with an upper end secured to the upper housing **130** and the product outlet **142** positioned a bottom end of the lower housing **132**. As such, agricultural product can be dispensed from the metering device **106** via the product outlet **142** of the lower housing **132**.

[0030] Embodiments provide for the metering devices **106** to convey various types of agricultural products. To facilitate such conveyances, the metering assembly **134** of the metering device **106** may comprise a plurality of metering wheels (e.g., metering wheel **143**), each being particularly configured to convey a particular type of agricultural product. As noted previously, the metering assembly **134** may be removable from the upper and lower housings **130**, **132**.

[0031] The metering assembly **134** may comprise a sub-shaft **146**, the metering wheels **143** positioned on the sub-shaft **146**, and a divider **147** positioned on the sub-shaft **146** between the metering wheels **143**. The sub-shaft **146** may comprise an elongated, hollow cylinder with an interior passageway having a surface shaped to conform to an exterior surface of a meter shaft discussed in more detail below. To facilitate rotation of the metering wheels **143**, an exterior surface of the sub-shaft **146** may be formed with one or more longitudinally-extending grooves or keyways **148** to secure the metering wheels **143** onto the sub-shaft **146** such that rotation of the sub-shaft **146** will cause a corresponding rotation of the metering wheels **143**.

[0032] The metering wheels **143** may each comprise a hollow interior section and a fluted exterior section. The interior section may include one or more protrusions or keys **149**, which are configured to be received in the grooves of the sub-shaft **146** when aligned. As such, the metering wheels **143** may be slid onto the sub-shaft **146** and secured in place via engagement between the keys **149** (of the metering wheels **143**) and the keyways **148** (of the sub-shaft **146**), such that rotation of the sub-shaft **146** will cause a corresponding rotation of the metering wheels **143**. The exterior sections of each of the metering wheels **143** may include a number of flutes **144** or concave grooves within which agricultural product (e.g., seed or treatment) can be received or captured for rotation through the metering device **106**. The size of the flutes **144** of each metering wheel **143** can vary depending on the type and size of the agricultural product intended to be processed by the metering device **106**.

[0033] In addition to the metering assembly **134**, certain embodiments of the present invention will provide for the metering device **106** to include gate assembly **136**. The gate assembly **136** may comprise a product gate **154** having an elongated, arcuate gate valve **158**. When installed within the metering device **106** during operation, the gate valve **158** will be spaced below the metering wheels **143** so as to provide a product channel **159** through which agricultural product can be conveyed by the metering wheels **143**. The size of the channel **159** can be adjusted by adjusting a position of the gate valve **158** with respect to the metering wheels **143**. For instance, shifting the gate valve **158** downward will create a larger channel **159**. Such a larger channel **159** may be preferable when using the metering device **106** to dispense relatively large agricultural products from the bin **104** (e.g., large seeds). Alternatively, shifting the gate valve **158** upward towards the metering wheels **143** will create a smaller channel **159**. Such a smaller channel may be preferable when using the metering device **106** to dispense relatively small agricultural products from the bin **104** (e.g., fine seeds). Ensuring the appropriate size of channel **159** for a given agricultural product will ensure consistent and accurate flow of such agricultural product through the metering device **106**. In some embodiments, it may be preferable to reduce the product channel **159** to a minimum such that the gate valve **158** is forced upward into contact with the metering wheels **143**.

[0034] As will be described in more detail below, the position of the gate valve **158** (and thus the size of the channel **159**) can be shifted by an adjustment shaft **160**, which can extend through each of the metering devices **106** of the drill **100**. Specifically, the adjustment shaft **160** may extend through each of the gate assemblies **136** of the metering devices **106**, such that rotation of the adjustment shaft **160** will cause a corresponding adjustment to the positions of each of the product gates **154**, and particularly to the gate valve **158**, with respect to the metering assemblies **134**. The adjustment shaft **160** can be rotated by various components or methods. For instance, the adjustment shaft **160** may be connected to a handle or lever **161** (FIG. **2**), which can be manually adjusted by an operator of the drill **100**. The lever **161** may be securely held in various positions, which correspond with the adjustment shaft **160** and/or the gate valve **158** being securely held in various positions. As was noted above, the gate valve **158** being positioned at various positions (e.g., further away from or closer to the metering wheels **143**) will provide for the channel **159** to have a correspondingly larger or smaller size. It is noted that a single lever **161** may be used to rotate the adjustment shaft **160**, such that the positions of each of the gate valves **158** of the metering devices **106** through which the adjustment shaft **160** extends can be simultaneously adjusted. As an alternative to the lever **161**, the adjustment shaft **160** may be connected to a motor or gear system, which can actuate the adjustment shaft **160** automatically or from a remote command provided by the operator of the drill **100** (e.g., from a cab of the tractor pulling the drill **100**).

[0035] Turning to FIGS. **4**, **6**, and **7**, the drive system **200** will now be described in more detail. The drive system **200** broadly comprises a meter shaft **202**, a mounting bracket **204**, a driven sprocket **206**, an idler sprocket **208**, a plurality of driver sprockets **210**, a chain **212**, and a plurality of electric motors **214**. The driven sprocket **206**, idler sprocket **208**, driver sprocket **210**, and chain **212** may be considered a drive train **226** drivably connecting the electric motors **214** to the meter shaft **202**. Alternative drive trains described in more detail below may be substituted for drive train **226** (see drive systems **300**, **400** below). The drive system **200** actuates the metering devices **106**, and more particularly, the metering wheels **143** of each metering device **106**.

[0036] The meter shaft **202** extends laterally between opposing ends of the drill **100** and is rotatably supported by the frame **102**. The meter shaft **202** also extends through the sub-shaft **146** of each metering device **106** and is rotationally entrained with each sub-shaft **146** so that the meter shaft **202** is drivably coupled with the metering wheels **143**. Furthermore, an end of the meter shaft **202** is drivably coupled with the driven sprocket **206**. To that end, the meter shaft **202** may extend through an opening in the mounting bracket **204**. In one embodiment, the meter shaft **202** is rectilinear in cross section. In a particular embodiment, the meter shaft **202** has a square cross section.

[0037] The driven sprocket **206** is rotatably positioned near the mounting bracket **204** on the end of the meter shaft **202** in alignment with the idler sprocket **208** and the driver sprockets **210**. The driven sprocket **206** is drivably coupled with the meter shaft **202** and includes a plurality of teeth **216** configured to engage the chain **212**. In one embodiment, the driven sprocket **206** includes a larger number of teeth than the driver sprockets **210** so that the driven sprocket **206** rotates slower than the driver sprockets **210** and with a relatively high torque. In one embodiment, the driven sprocket includes twenty-eight teeth.

[0038] The idler sprocket **208** is rotatably mounted on the mounting bracket **204** between the driven sprocket **206** and the driver sprockets **210** and may be adjustably attached to the mounting bracket **204** via slot **218**. The idler sprocket **208** takes up slack in the chain **212** to keep the chain **212** engaged with the driven sprocket **206** and the driver sprockets **210** and to prevent chain vibration. In one embodiment, the idler sprocket **208** includes **17** teeth-however, it is not particularly important how many teeth the idler sprocket **208** has because this number no bearing on speed or torque of the drive system **200**.

[0039] The driver sprockets **210** are rotatably positioned adjacent the mounting bracket **204** in spaced relation with the driven sprocket **206**. The driver sprockets **210** are also aligned with the driven sprocket **206** and the idler sprocket **208**. Each driver sprocket **210** is drivably coupled with one of the electric motors **214**. The driver sprockets **210** include teeth **222** configured to engage the chain **212**. In one embodiment, the driver sprockets **210** include a smaller number of teeth than the driven sprocket **206** so that the driver sprockets **210** rotate faster than the driven sprocket **206** and with relatively low torque. In one embodiment, each driver sprocket **210** includes twelve teeth. Alternatively, the driver sprockets **210** may have different numbers of teeth relative to each other and/or may be driven by motors having different operating characteristics. The benefits of such an arrangement will be described in more detail below.

[0040] The chain **212** drivably connects the driven sprocket **206**, the idler sprocket **208**, and the driver sprockets **210** so that rotation of the driver sprockets **210** effects rotation of the idler sprocket **208** and the driver sprockets **210**. To that end, the chain **212** includes a plurality of interconnected links **224** that engage teeth on the driven sprocket **206**, the idler sprocket **208**, and the driver sprockets **210**. Alternatively, in the case of pulleys used instead of sprockets, a belt or timing belt may be used to drivably connect the pulleys together.

[0041] The electric motors **214** are attached to the mounting bracket **204** and each includes a shaft **228** drivably connected to one of the driver sprockets **210**. The electric motors **214** may be substantially identical or may have different operating characteristics from each other. Such operating characteristics may include position, speed, torque, electrical current, voltage, direction of rotation, or the like. The electric motors **214** are communicatively connected to a control system (described below) for receiving control signals therefrom. To that end, one of the electric motors **214** may be designated a leader motor and the other electric motor **214** may be designated a follower motor, the significance of which will be described in more detail below. The electric motors **214** may be coupled with a clutch mechanism such as an over-running clutch for selectively disengaging the electric motors **214** from the metering devices **106**.

[0042] The control system is communicatively connected to the electric motors **214** (and to at least the lead electric motor) and is configured to transmit command signals thereto. The control system may be or may include an onboard controller, computer, or processor. Alternatively, the control system may be a remote computing device or computer system configured to communicate wirelessly with the electric motors **214**. Regardless, the control system may use pulse width modulation (PWM), control area network (CAN) communication, or the like for activating the leader motor.

[0043] The control system may implement aspects of the present invention with one or more computer programs stored in or on computer-readable medium residing on or accessible by the processing elements of the control system. Each computer program preferably comprises an ordered listing of executable instructions for implementing logical functions in the processing elements. Each computer program can be embodied in any non-transitory computer-readable medium for use by or in connection with an instruction execution system, apparatus, or device, such as a computer-based system, processor-containing system, or other system that can fetch the instructions from the instruction execution system, apparatus, or device, and execute the instructions. In the context of this application, a "computer-readable medium" can be any non-transitory means that can store the program for use by or in connection with the instructions execution system, apparatus, or device. The computer-readable medium can be, for example, an electronic, magnetic, optical, electro-magnetic, infrared, or semi-conductor system, apparatus, or device. More specifically, examples of the computer-readable medium may include an electrical connection having one or more wires, a portable computer diskette, a random access memory (RAM), a read-only memory (ROM), an erasable, programmable, read-only memory (EPROM or Flash memory), an optical fiber, and a portable compact disk read-only memory (CDROM).

[0044] The leader motor may be configured to receive a command signal from the control system to achieve a first operating state (e.g., a target setting or setpoint). The follower motor may be configured to achieve a second operating state based on the first operating state. In one embodiment, the second operating state may be substantially equal to the first operating state. The operating states may be a speed, a position, a torque, an electric current, a voltage, a direction of rotation, or the like. For example, the electric motors **214** may be synchronized so that the follower motor maintains a precise position and speed matching the leader motor so that each electric motor **214** contributes a portion of the power required to turn the meter shaft **202**.

[0045] The control system is configured to receive a feedback signal or signals representing current operating states of one or both of the electric motors **214** thus completing a control loop. The control system may be further configured to transmit additional command signals to the leader motor either in furtherance of achieving the first operating state or to proceed to subsequent operating states. For example, the control system may be configured to ramp up the output of the leader motor and hence the follower motor until the electric motors **214** reach a planting speed and later ramp down the output of the leader motor and hence the follower motor until the electric motors **214** stop upon completion of a planting operation.

[0046] The control system may also be configured to switch which of the electric motors **214** is a leader motor and which of the electric motors **214** is a follower motor. This may be useful for diagnostic purposes. Similarly, the control system may deactivate one of the electric motors **214** (e.g., the follower motor) such that only one of the electric motors **214** is active. This may be particularly useful when only one of the electric motors **214** is needed to power the metering devices **134**. In the same vein, the control system may activate the deactivated electric motor **214** when additional power is needed. In addition, the control system may receive user inputs, configuration information, and other programming for altering performance of the electric motors **214**. In this way, the electric motors **214** may be reconfigurable via user inputs.

[0047] In operation, agricultural product will flow from the bin **104** into the interior space of the metering device **106** via the product inlet **137** (e.g., through one of the open product doors). The agricultural product is propelled through the interior space of the metering device **106** via rotation of the metering wheels **143** of the metering assembly **134**. Specifically, agricultural product will be captured by the flutes **144** on the metering wheels **143**, such that rotation of the metering wheels **143** will convey the agricultural product counterclockwise (when viewing FIG. **5**), around the interior space of the metering device **106** to the product gate **154**. At such point, the metering wheels **143** continue to convey the agricultural product through the channel **159** presented between the metering wheels **143** and the gate valve **158** of the product gate **154** until the agricultural product passes over the gate valve **158** and falls out of the metering device **106** through the product outlet **142**. As should be apparent, the size of the channel **159** can be selected to permit a particular type and size of agricultural product to be efficiently conveyed through the metering device **106**. For larger agricultural product, the size of the channel **159** can be increased by positioning the gate valve **158** further away from the metering wheels **143** by rotating the gate assembly **136** via rotation of the adjustment shaft **160**. Alternatively, for smaller agricultural product, the size of the channel **159** can be decreased by positioning the gate valve **158** closer to the metering wheels **143** by rotating the gate assembly **136** via rotation of the adjustment shaft **160**.

[0048] The metering wheels **143** of the metering devices **106** are driven by the drive system **200** as controlled by the control system. Specifically, the control system transmits a control signal to the leader motor to achieve a first operating state (e.g., a target setting or setpoint). As described above, the follower motor achieves a second operating state based on the first operating state. In one embodiment, the second operating state may be substantially equal to the first operating state. The operating states may be a speed, a position, a torque, an electric current, a voltage, a direction of rotation, or the like. For example, the electric motors **214** may be synchronized so that the follower motor maintains a precise position and speed matching the leader motor so that each electric motor **214** contributes a portion of the power required to turn the meter shaft **202**. The electric motors **214** thereby turn the driver sprockets **210**, which actuate the driven sprocket **206** and hence the meter shaft **202** via the chain **212**. The meter shaft **202** in turn actuates the metering wheels **143** so that agricultural product is metered through the metering devices **106**.

[0049] The control system then receives a feedback signal or signals representing current operating states of one or both of the electric motors **214** thus completing a control loop. The control system then transmits additional command signals to the leader motor either in furtherance of achieving the first operating state or to proceed to subsequent operating states. For example, the control system may ramp up the output of the leader motor and hence the follower motor until the electric motors **214** reach a planting speed and later ramp down the output of the leader motor and hence the follower motor until the electric motors **214** stop upon completion of a planting operation.

[0050] A method of planting seeds will now be described in more detail. First, a common meter shaft may be connected to a plurality of individual seed meters so that rotation of the common meter shaft causes rotation of the plurality of individual seed meters. Next the common meter shaft may be rotatably driven in a first operating state where n individual electric motors power rotation of the common meter shaft. This may include transmitting a control signal to a first one of the n individual electric motors, with the control signal representing a command to achieve a setpoint. The first one of the n individual electric motors may thereby operate as a leader motor.

[0051] The common meter shaft may then be rotatably driven in a second operating state where m individual electric motors power rotation of the common meter shaft, where m is at least one more than n. In other words, at least one additional electric motor may be activated to drive the common motor shaft when additional power is needed or desired. The at least one additional electric motor may operate as a follower motor so as to achieve a second operating state corresponding to the first operating state. The leader and follower motors may have different operating specifications from each other so that the follower motor is operated at a different power output than the leader motor.

[0052] Turning to FIG. **8**, a drive system **300** constructed in accordance with another embodiment of the invention will now be described. The drive system **300** broadly comprises a meter shaft **302**, one or more mounting brackets (not shown), a driven gear **306**, a plurality of driver gears **310**, and a plurality of electric motors **314**. The driver gears **310** and driven gear **306** form drive train **326** drivably connecting the electric motors **314** to the meter shaft **302**. The drive system **300** differs from the above-described drive system **200** in that unlike the use of indirect chain or belt drive, each one of the driver gears **310** is configured to directly engage the driven gear **306**.

[0053] The meter shaft **302** extends laterally between opposing ends of a drill and is supported by the drill's frame. The meter shaft **302** also extends through the sub-shafts of a plurality of metering devices (not shown) and is rotationally entrained with the sub-shafts so that the meter shaft **302** is drivably coupled with metering wheels of the metering devices. Furthermore, the meter shaft **302** is drivably coupled with the driven gear **306**.

[0054] The driven gear **306** is drivably coupled with the meter shaft **302** and includes a plurality of teeth **316** configured to engage teeth of the driver gears **310**. In one embodiment, the driven gear **306** includes a larger number of teeth than the driver gears **310** so that the driven gear **306** rotates slower than the driver gears **310** and with relatively high torque.

[0055] The driver gears **310** are aligned and in driving engagement with the driven gear **306**. Each of the driver gears **310** is also drivably coupled with one of the electric motors **314**. The driver gears **310** include teeth **322** configured to engage teeth of the driven gear **306**. In one embodiment, each of the driver gears **310** includes a smaller number of teeth than the driven gear **306** so that the driver gears **310** rotate faster than the driven gear **306** and with relatively low torque. Alternatively, the driver gears **310** may have different numbers of teeth relative to each other and/or may be driven by motors having different operating characteristics.

[0056] The electric motors **314** each include a shaft **328** drivably connected to one of the driver gears **310**. The electric motors **314** may be substantially identical or may have different operating characteristics from each other. Such operating characteristics may include position, speed, torque, electrical current, voltage, direction of rotation, or the like. The electric motors **314** are communicatively connected to a control system for receiving control signals therefrom. To that end, one of the electric motors **314** may be designated a leader motor and the other electric motor **314** may be designated a follower motor as described above. The electric motors **314** may be coupled with a clutch mechanism such as an over-running clutch for selectively disengaging the electric motors **314** from the metering devices.

[0057] The above-described drive system **300** operates as follows. The leader motor receives a control signal to achieve a first operating state (e.g., a target setting or setpoint). The follower motor achieves a second operating state based on the first operating state. In one embodiment, the second operating state may be substantially equal to the first operating state. The operating states may be a speed, a position, a torque, an electric current, a voltage, a direction of rotation, or the like. For example, the electric motors **314** may be synchronized so that the follower motor maintains a precise position and speed matching the leader motor so that each electric motor **314** contributes a portion of the power required to turn the meter shaft **302**. The electric motors **314** thereby turn the driver gears **310**, which actuate the driven gear **306** and hence the meter shaft **302**. The meter shaft **302** in turn actuates metering wheels of corresponding metering devices so that agricultural product is metered through the metering devices.

[0058] Turning to FIG. **9**, a drive system **400** constructed in accordance with another embodiment of the invention will now be described. The drive system **400** broadly comprises a meter shaft **402**, one or more mounting brackets (not shown), a plurality of driven gears **406**, a plurality of driver gears **410**, and a plurality of electric motors **414**.

[0059] The driver gears **410** and driven gears **406** form drive train **426** drivably connecting the electric motors **414** to the meter shaft **402**. However, the drive train **426** may instead have driven sprockets and driver sprockets drivably connected by chains or driven pulleys and driver pulleys drivably connected by belts. The drive system **400** differs from the above-described drive systems **200**, **300** in that the electric motors **414** are drivably connected to the meter shaft **402** at different axial locations on the meter shaft **402** from each other.

[0060] The meter shaft **402** extends laterally between opposing ends of a drill and is supported by the drill's frame. The meter shaft **402** also extends through the sub-shafts of a plurality of metering devices (not shown) and is rotationally entrained with the sub-shafts so that the meter shaft **402** is drivably coupled with metering wheels of the metering devices. Furthermore, the meter shaft **402** is drivably coupled with the driven gears **406** at different axial locations on the meter shaft **402**. In one embodiment, the meter shaft **402** is rectilinear in cross section. In a particular embodiment, the meter shaft **402** has a square cross section.

[0061] The driven gears **406** are drivably coupled with the meter shaft **402** and spaced apart from each other at different locations on the meter shaft **402**. Each driven gear **406** includes a plurality of teeth **416** configured to engage teeth of one of the driver gears **410**. In one embodiment, the driven gears **406** each include a larger number of teeth than the driver gears **410** so that the driven gears **406** rotate slower than the driver gears **410** and with relatively high torque.

[0062] Each of the driver gears **410** are aligned and in driving engagement with one of the driven gears **406**. Each of the driver gears **410** is also drivably coupled with one of the electric motors **414**. Each of the driver gears **410** includes teeth **422** configured to engage teeth of the driven gears **406**. In one embodiment, the driver gears **410** include a smaller number of teeth than the driven gears **406** so that the driver gears **410** rotate faster than the driven gears **406** and with relatively low torque. Alternatively, the driver gears **410** may have different numbers of teeth relative to each other and/or may be driven by motors having different operating characteristics.

[0063] The electric motors **414** each include a shaft **428** drivably connected to one of the driver gears **410**. The electric motors **414** may be substantially identical or may have different operating characteristics from each other. Such operating characteristics may include position, speed, torque, electrical current, voltage, direction of rotation, or the like. The electric motors **414** are communicatively connected to a control system for receiving control signals therefrom. To that end, one of the electric motors **414** may be designated a leader motor and the other electric motor **414** may be designated a follower motor as described above. At least one of the electric motors **414** may be coupled with a clutch mechanism such as an over-running clutch for selectively disengaging the at least one of the electric motors **414** from the metering devices, which may be useful at least for switching leader/follower roles of the electric motors **414**. The leader motor may or may not have a clutch. The clutch may be a frictional clutch or an overrunning clutch such as a wrap spring type clutch. The clutch may be a manually engaged cog drive or a removable pin which may be suitable for infrequent operation thereof.

[0064] The above-described drive system **400** operates as follows. The leader motor receives a control signal to achieve a first operating state (e.g., a target setting or setpoint). The follower motor achieves a second operating state based on the first operating state. In one embodiment, the second operating state may be substantially equal to the first operating state. The operating states may be a speed, a position, a torque, an electric current, a voltage, a direction of rotation, or the like. For example, the electric motors **414** may be synchronized so that the follower motor maintains a precise position and speed matching the leader motor so that each electric motor **414** contributes a portion of the power required to turn the meter shaft **402**. The electric motors **414** thereby turn the driver gears **410**, which actuate the driven gears **406** and hence the meter shaft **402**. The meter shaft **402** in turn actuates metering wheels of corresponding metering devices so that agricultural product is metered through the metering devices.

[0065] Although the invention has been described with reference to the one or more embodiments illustrated in the figures, it is understood that equivalents may be employed and substitutions made herein without departing from the scope of the invention as recited in the claims.

[0066] Having thus described one or more embodiments of the invention, what is claimed as new and desired to be protected by Letters Patent includes the following:

What is claimed is:

1. A drive system for delivering a plurality of seeds into a plurality of furrows via a plurality of seed meters, the drive system comprising:
- a meter shaft drivably connectable to the plurality of seed meters; and
- a plurality of electric motors drivably connected to the meter shaft for actuating the plurality of seed meters via the meter shaft.

2. The drive system of claim **1**, wherein at least two of the plurality of electric motors are drivably connected to the meter shaft at different axial locations on the meter shaft from each other.

3. The drive system of claim **1**, further comprising a drive train, wherein at least two of the plurality of electric motors are drivably connected to the meter shaft at one axial location on the meter shaft via the drive train.

4. The drive system of claim **3**, wherein the drive train includes a driven gear coupled to the meter shaft and a driver gear coupled to each one of the plurality of electric motors, each one of the driver gears being configured to directly engage the driven gear.

5. The drive system of claim **3**, wherein the drive train includes a driven sprocket coupled to the meter shaft, a driver sprocket coupled to each one of the plurality of electric motors, and a drive chain linking the drive sprockets to the driven sprocket.

6. The drive system of claim **1**, wherein at least one of the plurality of electric motors is drivably connected to the meter shaft via a clutch.

7. The drive system of claim **1**, wherein the plurality of electric motors includes two motors having different operating specifications from each other.

8. The drive system of claim **1**, wherein at least one of the plurality of electric motors is reconfigurable via a user input.

9. A seed drill for delivering a plurality of seeds into a plurality of furrows, the seed drill comprising:
- a drive system comprising:
  - a plurality of seed meters each configured to deliver some of the plurality of seeds into one of the plurality of furrows;
  - a meter shaft drivably connected to the plurality of seed meters; and
  - a plurality of electric motors drivably connected to the meter shaft for actuating the plurality of seed meters via the meter shaft, the plurality of electric motors including:
    - a first electric motor operating as a leader motor; and
    - a second electric motor operating as a follower motor; and
- a control system communicatively connected to the first electric motor and configured to transmit a command signal to the first electric motor for activating the first electric motor to achieve a first operating state.

10. The seed drill of claim **9**, wherein the second electric motor is configured to achieve a second operating state based on the first operating state.

11. The seed drill of claim **10**, wherein the second electric motor is synchronized with the first electric motor so that the second operating state is substantially equal to the first operating state.

12. The seed drill of claim **10**, wherein each of the first operating state and the second operating state are at least one of a speed, a torque, an electric current, a voltage, and a direction of rotation.

13. The seed drill of claim **9**, wherein the second electric motor is configured to be switched to operating as a leader motor.

14. The seed drill of claim **9**, wherein the first and second electric motors have different operating specifications from each other.

15. The seed drill of claim **9**, wherein one of the first and second electric motors is reconfigurable via a user input.

16. The seed drill of claim **9**, wherein the first and second electric motors are drivably connected to the meter shaft at different axial locations on the meter shaft from each other.

17. The seed drill of claim **9**, the drive system further comprising a drive train, wherein the first and second electric motors are drivably connected to the meter shaft at one axial location on the meter shaft via the drive train.

18. The seed drill of claim **17**, wherein the drive train includes a driven gear coupled to the meter shaft and a driver gear coupled to each one of the plurality of electric motors, each one of the driver gears being configured to directly engage the driven gear.

19. The seed drill of claim **17**, wherein the drive train includes a driven sprocket coupled to the meter shaft, a driver sprocket coupled to each one of the plurality of electric motors, and a drive chain linking the drive sprockets to the driven sprocket.

20. A seed drill for planting a plurality of seeds in a plurality of furrows, the seed drill comprising:

a drive system comprising:

a plurality of seed meters each configured to deliver some of the plurality of seeds into one of the plurality of furrows;

a meter shaft drivably connected to the plurality of seed meters;

a plurality of electric motors drivably connected to the meter shaft for actuating the plurality of seed meters via the meter shaft, the plurality of electric motors including:

a first electric motor operating as a leader motor; and

a second electric motor operating as a follower motor and configured to be switched to operating as a leader motor, the first and second electric motors having different operating specifications from each other; and

a drive train drivably connecting the first electric motor and the second electric motor to the meter shaft at one axial location on the meter shaft; and

a control system communicatively connected to the first electric motor and configured to transmit a command signal via pulse width modulation (PWM) control area network (CAN) communication to the first electric motor for activating the first electric motor to achieve a first operating state, the second electric motor being synchronized with the first electric motor to achieve a second operating state substantially equal to the first operating state, the first and second operating states being at least one of a speed, a torque, an electric current, a voltage, and a direction of rotation.

21. A method of planting seeds, the method comprising steps of:

(a) connecting a common meter shaft to a plurality of individual seed meters so that rotation of the common meter shaft causes rotation of the plurality of individual seed meters; and

(b) driving rotation of the common meter shaft via a plurality of individual electric motors.

22. The method of claim **21**, further comprising a step of:

(c) before step (b), driving rotation of the common meter shaft in a first operating state where n individual electric motors power rotation of the common meter shaft,

wherein step (b) includes driving rotation of the common meter shaft in a second operating state where m individual electric motors power rotation of the common meter shaft,

wherein m is at least one more than n.

23. The method of claim **22**, wherein said n individual electric motors includes a first electric motor, said m individual electric motors includes the first electric motor and a second electric motor, and the method further comprises a step of synchronizing the second electric motor to the first electric motor before step (b).

24. The method of claim **23**, wherein the first electric motor operates as a leader motor and the second electric motor operates as a follower motor in step (b), the method further comprising a step of transmitting a signal to the first electric motor representing a command to achieve a setpoint so that the first and second electric motors achieve the setpoint.

25. The method of claim **22**, wherein said n individual electric motors includes a first electric motor, said m individual electric motors includes the first electric motor and a second electric motor, the first electric motor and the second electric motor have different operating specifications from each other, and the method further comprises a step of operating the second electric motor at a different power output than the first electric motor.

* * * * *